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ORIGINAL ARTICLE

Transforming oncology clinical trial matching through neuro-symbolic, multi-agent AI and an oncology-specific knowledge graph: a prospective evaluation in 3804 patients

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Background: Clinical trial enrollment in oncology remains critically low, with fewer than 5% of eligible adults participating, in large part due to the complexity and labor intensity of eligibility screening. We prospectively evaluated a neuro-symbolic, multi-agent artificial intelligence (AI) platform integrating domain-specific large language model (LLM) agents, an oncology-specific knowledge graph, a real-time recommendation engine, and human-in-the-loop review to determine whether automated extraction and reasoning can safely improve trial identification, efficiency, and equity at scale.

Methods: Consecutive patients [$N = 3804$; Eastern Cooperative Oncology Group (ECOG) 0-2] balanced for cancer type incidence with metastatic or progressive malignancies were screened across a 12-month period. A multi-agent architecture—OncoAgents (LLM-based extraction and reasoning agents), OncoGraph (oncology knowledge graph), OncoRecommend (prioritization engine), and OncoSet (expert-curated corpus)—carried out automated data extraction, harmonization, and trial matching over 157 367 clinical pages (∼86.5 M tokens). Dual oncologists produced a gold standard of trial eligibility labels (Cohen's $\kappa = 0.92$). The primary unit of analysis was the patient—trial pair. Baselines included manual screening, GPT-4 zero-shot prompting, GPT-4 chain-of-thought, and frontier GPT-4o extraction/matching benchmarks. Outcomes included sensitivity, specificity, precision, F1 score, calibration of eligibility confidence scores, time-to-recommendation, fairness across demographic subgroups, and operational burden.

Results: The multi-agent neuro-symbolic system achieved an F1 score of 0.82 (95% confidence interval 0.81-0.83). In comparison, the GPT-4 zero-shot baseline achieved an F1 of 0.47, and the GPT-4 chain-of-thought baseline achieved an F1 of 0.67. Per-patient screening time decreased from a median of 120 min (manual review) to ∼30 min total (15 min automated processing + 15 min clinical review). Across the cohort, the system processed 157 000 pages, screened 23 912 candidate patient—trial pairs, and produced 17 912 oncologist-confirmed matches, with median time-to-recommendation <7 days. No demographic subgroup exceeded a 10-percentage point F1 gap; the largest observed difference was ∼7 points between white and black/African American patients. Ablation experiments showed that both knowledge graph grounding and multi-agent decomposition contributed materially to performance and efficiency. Eligibility confidence scores exhibited reasonable calibration in the clinically relevant operating range.

Conclusions: A neuro-symbolic, multi-agent architecture that couples LLM-based extraction with ontology-grounded, deterministic eligibility reasoning improved the accuracy, throughput, and timeliness of oncology clinical trial matching versus LLM-only baselines, while preserving clinician oversight and maintaining modest subgroup performance gaps. These results support scalable, equity-aware deployment of AI-assisted trial screening in routine oncology practice.

Key words: neuro-symbolic AI, clinical trial matching, knowledge graph, multi-agent systems, large language models (LLMs), oncology decision support

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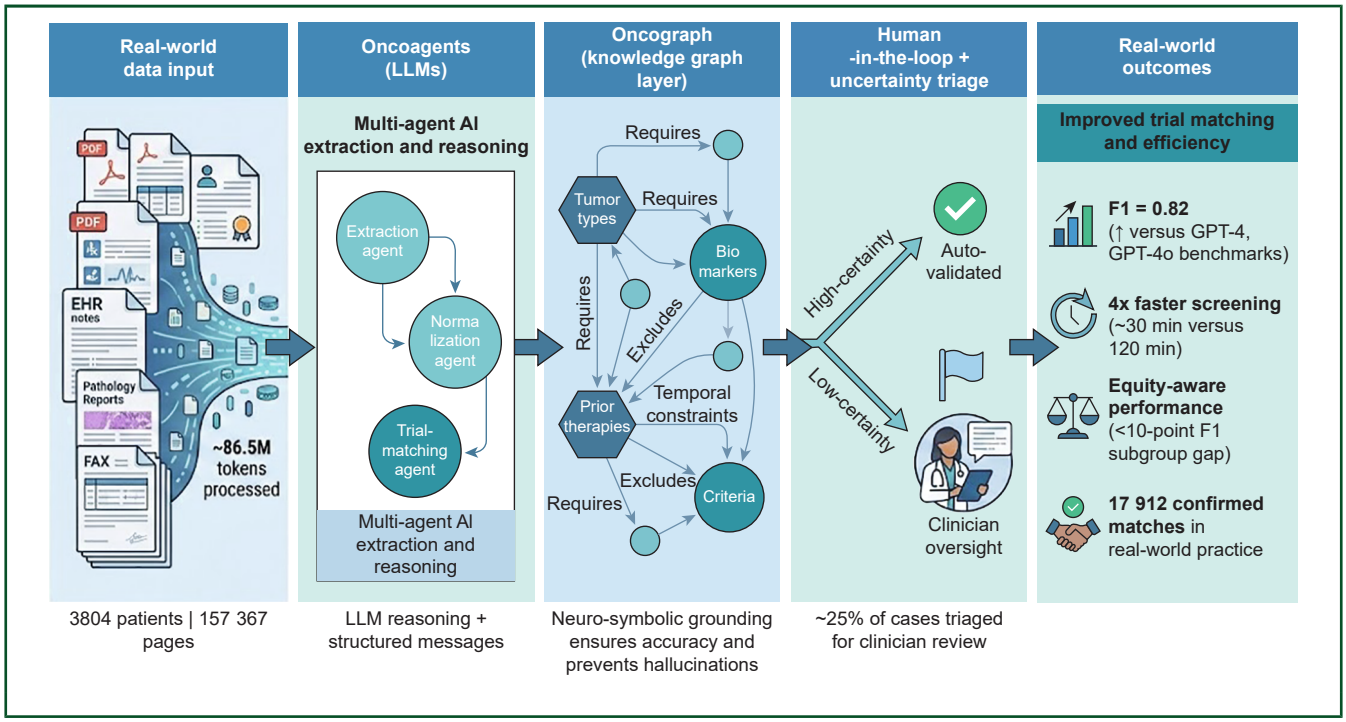
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INTRODUCTION

The persistent crisis of oncology trial enrollment

Clinical trials drive oncology progress, yet only 3%-5% of adult cancer patients enroll in research studies.[1,2] Enrollment is limited less by trial availability than by the effort of mapping fragmented longitudinal records to complex,

GRAPHICAL ABSTRACT



exception-rich eligibility criteria that are increasingly genomic, temporal, and organ-function specific.

Manual screening typically takes ∼120 min per patient and requires multi-year chart review across disparate systems; in high-volume (especially community) practice this is unsustainable and contributes to missed trial opportunities.

From deterministic rules to neuro-symbolic artificial intelligence

Trial-matching tools have progressed from rule/keyword systems to natural language processing (NLP), retrieval-augmented pipelines, and zero-shot large language models (LLMs),[3,4] but the hardest part of eligibility screening is still not simply finding relevant text. It is translating free-text oncology criteria into executable, auditable logic that can be applied to fragmented longitudinal records containing negation, temporality, laboratory thresholds, and exceptions.

This is the specific gap addressed here. Prior artificial intelligence (AI)-based matching systems have shown promise for candidate retrieval or narrow-scope matching, but fewer studies have prospectively evaluated broad oncology workflows that combine multimodal electronic health record (EHR) extraction, ontology normalization, deterministic temporal reasoning, and clinician review within one framework. Even frontier LLMs (e.g. GPT-4o) remain probabilistic and can hallucinate criteria or mishandle timing, making direct end-to-end deployment risky.

We therefore evaluated a neuro-symbolic architecture in which LLM agents carry out narrative extraction and normalization, while an oncology-specific knowledge graph executes explicit inclusion, exclusion, and exception logic. This differs from LLM-only matching systems because recommendations are not returned directly from free text; they must first be mapped to canonical concepts and pass graph-based checks for numeric, Boolean, and temporal consistency.[5-7]

Purpose of this study

We tested whether a multi-agent neuro-symbolic system grounded in an oncology knowledge graph and evaluated against a physician gold standard could:

- Improve identification accuracy versus manual screening and LLM-only baselines.
- Reduce per-patient screening time and operational burden.
- Preserve clinician oversight via human-in-the-loop review and concise rationales.
- Assess equity across demographic subgroups.
- Outperform zero-shot, chain-of-thought (CoT), and frontier GPT-4o baselines in extraction and matching.

This full report extends a previously presented conference abstract[8] by providing full methodological detail, ablation, calibration, and fairness analyses, together with CONSORT-AI-aligned reporting[9] (Supplementary Table S1, available at https://doi.org/10.1016/j.esmorw.2026.100706).

METHODS

Study design and cohort

From January to December 2024, we prospectively evaluated 3804 consecutive patients with metastatic or progressive solid tumors [Eastern Cooperative Oncology Group (ECOG) 0-2] across US academic—community sites. Screening eligibility was intentionally broad to mirror routine practice (active/progressing disease and candidates for additional therapy), minimizing selection bias; patients were screened at referral and re-screened as new imaging, pathology, or genomic results accrued (Supplementary Table S2, available at https://doi.org/10.1016/j.esmorw.2026.100706).

OncoSet served only for development and few-shot prompting; the prospective cohort was fully held out from prompt engineering and calibration. The trial library included open interventional phase I-III trials recruiting locally, plus curated nationwide referral trials for rare molecular subtypes. Trial availability was updated regularly during the study period.

Gold standard annotation

Two board-certified medical oncologists independently reviewed each patient's complete EHR (pathology, imaging, clinical notes, genomics, and treatment history) to label eligibility for relevant candidate trials (see Baselines and comparative models) using a predefined interpretation protocol.

κ Inter-rater reliability was high (Cohen's $\kappa = 0.92$); discrepancies were resolved by consensus. This dual-oncologist dataset served as the gold standard for evaluation and as a calibration corpus for error analysis.

To prioritize safety, if a mandatory criterion (e.g. a required biomarker) was undocumented, patients were labeled Not Eligible for that constraint. Ambiguous temporal references without dates were flagged as Requires Clinician Judgment and excluded from the binary F1 calculation.

Multi-agent neuro-symbolic AI system

The platform comprises OncoAgents, OncoGraph, OncoRecommend, and OncoSet (Figure 1), implementing a neuro-symbolic architecture in which probabilistic LLM agents operate over deterministic oncology graph logic.

OncoAgents: specialized LLM subsystems. OncoAgents are domain-tuned LLM agents for extraction, normalization, and matching:

- Data extraction agent: performs optical character recognition (OCR)/layout parsing and NLP to extract tumor type/stage, prior therapy, biomarkers, ECOG status, and key comorbidities, emitting structured JSON ('OncoSet feature vector').
- Eligibility agent: normalizes extracted features to canonical ontologies, resolving negation and temporality (e.g. active versus remote), producing a graph-aligned feature set.
- Trial-matching agent: evaluates the structured patient representation against OncoGraph inclusion/exclusion logic, returning ranked candidate trials with explicit reasoning.

Agents exchange structured messages that include CoT reasoning to detect conflicts and estimate uncertainty. Clinicians see a short rationale linked to supporting EHR passages, enabling rapid verification and audit while keeping raw notes within the secure environment.

OncoGraph: oncology knowledge graph and criteria parsing. OncoGraph is a curated oncology knowledge graph that encodes trial criteria, disease and biomarker taxonomies, prior therapies, and organ-function requirements as nodes and edges.

- Nodes represent tumor subtypes, genomic alterations, biomarkers, drug classes, comorbidities, and trial entities.
- Edges capture relationships such as 'requires', 'excludes', 'contraindicated-with', and 'progressed-after'.

OncoGraph is aligned with:

- National Cancer Institute (NCI) Thesaurus for detailed oncology disease and biomarker concepts.
- Systematized Nomenclature of Medicine—Clinical Terms (SNOMED CT) for clinical findings and comorbid conditions.
- MeSH for interventions and pharmacologic classifications.

Knowledge graph curation and temporal logic. Trials from ClinicalTrials.gov are ingested monthly. Free-text criteria are parsed using pattern rules plus LLM-assisted extraction, then represented as atomic clauses that are mapped to ontology terms. Curators store each clause at three levels: the literal trial text, a normalized concept representation, and an executable graph rule. Temporal constraints are represented with event nodes (e.g. treatment start/end, progression, steroid exposure) and interval windows anchored to a screening date; negations and exceptions are encoded as explicit exclusion or override edges then mapped to ontology terms.

Laboratory thresholds are normalized to common units with assay notes when needed; ambiguous criteria are flagged for curator review. Protocol amendments are reconciled conservatively, and unresolved ambiguity is routed to Requires Clinician Judgment rather than permissive auto-matching. All OncoGraph updates are version-controlled to support drift analysis (Boxes 1 and 2).

OncoRecommend: real-time prioritization engine. OncoRecommend orchestrates OncoAgents and OncoGraph to:

- Ingest new or updated EHR data.
- Apply eligibility constraints via OncoGraph queries.
- Prioritize trials by clinical relevance and degree-of-fit.
- Return ranked trial lists with confidence scores and brief rationales.

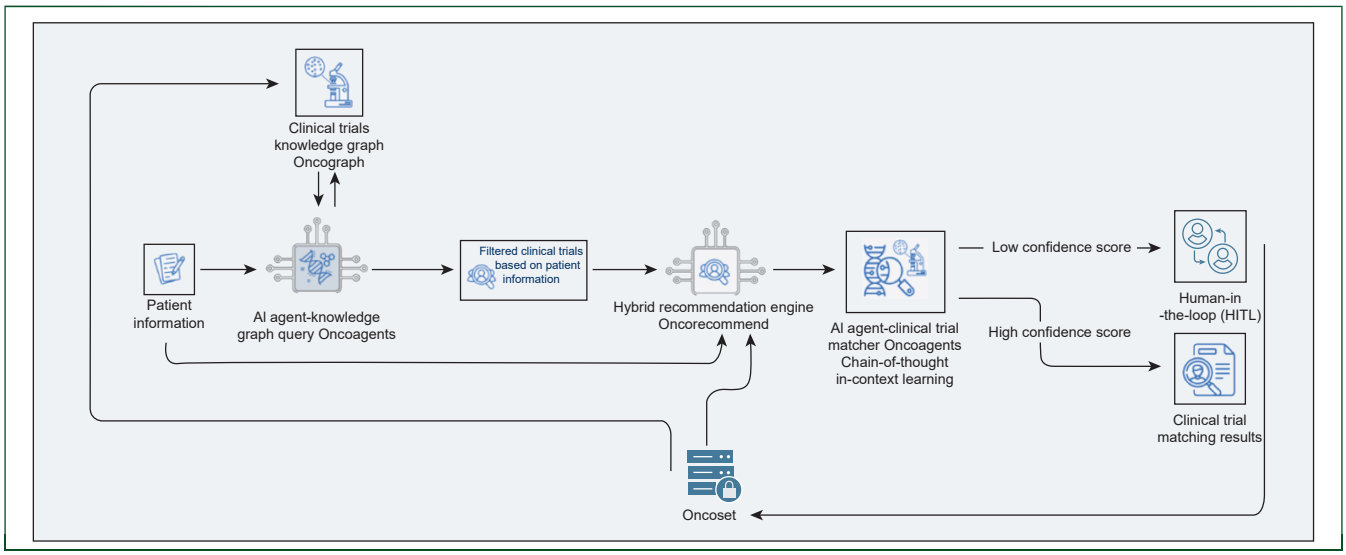


Figure 1. **High-level architecture of the neuro-symbolic, multi-agent AI platform, showing OncoAgents, OncoGraph, and human-in-the-loop review.**

Low-confidence or conflicting evidence (∼15% of patient—trial pairs) triggers prioritized human review.

OncoSet: expert-curated corpus. OncoSet is an expert-curated dataset (>2000 annotated records, 14 000+ trials, >50 tumor subtypes) used to train and calibrate OncoAgents and OncoGraph alignment. The 3804-patient prospective cohort served as a held-out evaluation set.

Data volume and processing

Across 3804 patients, the platform processed:

- 157 367 clinical document pages.
- Approximately 86.5 million text tokens.
- Mixed modalities (structured notes plus scanned PDFs/faxes, pathology reports, and imaging impressions).

This volume reflects real-world documentation heterogeneity and enabled robust evaluation of extraction performance.

Baselines and comparative models

We compared the multi-agent system with four baselines:

- Manual screening: conventional coordinator/oncologist eligibility review (workflow reference, not a statistical F1 comparator).
- GPT-4 zero-shot: single zero-shot prompt using raw patient summaries and trial text, without graph structure.
- GPT-4 CoT: two-step prompt: extract key features, then reason step-by-step over criteria.
- GPT-4o frontier benchmark: Frontier multimodal model on identical extraction and matching tasks, using the same structured prompts where applicable.

Eighty percent of cases were used for training and internal validation (10% split), with 20% held out for testing. Prompt templates, token limits, and temperature settings are detailed in the Supplementary Methods, available at https://doi.org/10.1016/j.esmorw.2026.100706; all LLM baselines were evaluated on the same filtered candidate

Box 1. **End-to-end example with temporal logic, exception handling, ontology mapping, curation, and executable evaluation**

Worked example (raw eligibility text): 'Inclusion: metastatic urothelial carcinoma with radiographic progression during or after platinum-based chemotherapy; last dose of systemic therapy must be ≥14 days before cycle 1 day 1. Exclusion: active pneumonitis requiring steroids within the past 12 months. Prior PD-1/PD-L1 therapy is allowed if discontinued ≥28 days before cycle 1 day 1.'

Parsing and ontology mapping: the clause parser decomposes this into requires(disease = metastatic urothelial carcinoma), requires(progressed-after(platinum-based chemotherapy)), requires(interval[last_systemic_therapy_end, C1D1] ≥ 14 days), excludes(active pneumonitis AND steroid_use_within_12_months), and exception(allows prior PD-1/PD-L1 therapy if interval[last_checkpoint_inhibitor_end, C1D1] ≥ 28 days). Drug mentions such as cisplatin are mapped to Platinum Agent in the NCI Thesaurus; pneumonitis is mapped to a SNOMED CT disorder concept with status attributes such as active or resolved.

Patient evidence: a note stating 'cisplatin/gemcitabine stopped 2023-03-10 for enlarging hepatic metastases; pembrolizumab ended 2023-08-01; radiation pneumonitis in 2022 resolved, no steroids since 2022-09-15; planned screening date 2024-02-01' is converted into event nodes for prior platinum exposure, radiographic progression, prior PD-1 therapy, pneumonitis status, and last steroid use.

Curated executable rule and evaluation: OncoGraph evaluates progressed-after(Patient, Platinum_Based_Chemotherapy) = true; interval(2023-03-10, 2024-02-01) ≥ 14 days = true; interval(2023-08-01, 2024-02-01) ≥ 28 days = true; active_pneumonitis = false; steroid_use_within_12_months = false. The patient is therefore marked Eligible. If the pneumonitis status were unclear or the steroid stop date could not be verified, the pair would not be auto-approved; it would be triaged to Requires Clinician Judgment.

C1D1, cycle 1, day 1; PD-1, programmed cell death protein 1; PD-L1, programmed death-ligand 1; SNOMED CT, Systematized Nomenclature of Medicine—Clinical Terms.

Box 2. Example of inter-agent reasoning and knowledge graph grounding

Scenario: A trial requires 'progression on prior platinum-based chemotherapy.' The patient's note states: '...discontinued Cisplatin/Gemcitabine in March 2023 due to growing hepatic mets...'
Step 1: Data extraction agent (unstructured to structured)

- Input: Raw clinical note segment.
- Output (JSON):

```
JSON
{
 'treatment_line': {
 'agents': ['Cisplatin', 'Gemcitabine'],
 'start_date': '2023-01-15',
 'end_date': '2023-03-10',
 'outcome': 'progression',
 'context': 'growing hepatic mets'
 }
}
```

Step 2: OncoGraph query (structured to knowledge grounding)

- Query: Is ['Cisplatin', 'Gemcitabine'] subclass of Platinum_Based_Chemotherapy?
- Result: TRUE (Cisplatin to Platinum Agent).
- Query: Does outcome: 'progression' satisfy refractory/relapsed criteria?
- Result: TRUE.Step 3: Trial-matching agent (reasoning to decision)
- Logic: Patient received verified platinum agent AND experienced progression to inclusion criteria met.
- Final output: ELIGIBLE

trial sets. For GPT-4o we also evaluated a retrieval-augmented variant.

Evaluation units and labeling strategy

The primary unit of analysis was the 'patient—trial pair'. Candidate trials were pre-filtered using high-level attributes (site/histology, metastatic status, ECOG, key biomarker flags), then the gold standard labeled each patient—trial pair as:

- Eligible: all automatable criteria satisfied with no disqualifying criterion.
- Not eligible: at least one clearly disqualifying criterion present.
- Requires Clinician Judgment: eligibility depends on nuanced interpretation (e.g. borderline imaging or complex comorbidity trade-offs).

We computed sensitivity, specificity, precision, and F1 over pairs labeled eligible or not eligible. Pairs labeled Requires Clinician Judgment were described separately but excluded from the binary F1 calculation to mirror real-world clinician decision-making.

Fairness analysis

We evaluated subgroup performance by age, sex, race, ethnicity, computing sensitivity, specificity, precision, and F1 at the patient—trial level (Figure 2).

A pre-specified absolute F1 gap >10 points between any two groups triggered a fairness alert. Subgroup sizes and 95% confidence intervals (nonparametric bootstrap) are reported in Supplementary Table S5, available at https://doi.org/10.1016/j.esmorw.2026.100706.

Deployment, data governance, and compute footprint

The system ran in the health networks' secure environment under a Health Insurance Portability and Accountability Act (HIPAA)-aligned business associate agreement and patient consent, processing protected health information on-premises or in a role-controlled virtual private cloud.

Intermediate agent messages and CoT explanations remained within this environment; clinician-facing rationales linked to EHR evidence rather than reproducing long text.

Median model-processed volume was ~23 000 tokens per patient, corresponding to an estimated low single-digit

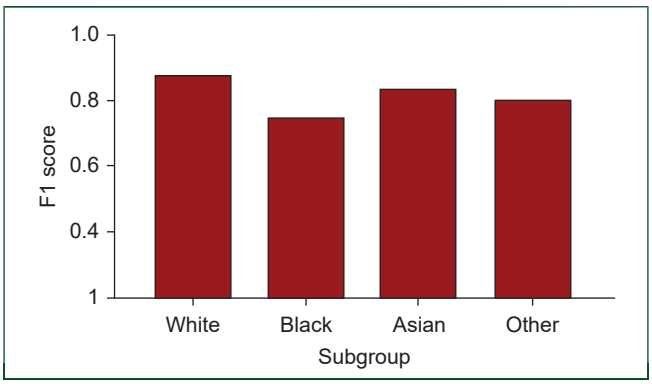


Figure 2. Illustrative subgroup F1 performance by race.

dollar compute cost; costs vary by model choice and re-screening frequency.

RESULTS

Overall performance

Table 1 summarizes the main performance metrics at the patient—trial level on the held-out test set.

The neuro-symbolic system achieved balanced sensitivity and specificity (∼0.84), precision 0.81, and F1 0.8246 (95% confidence interval ∼0.81-0.85). The pattern across Table 1 is informative: performance improved from zero-shot prompting to CoT prompting and then again to the full system, suggesting that prompt engineering alone explained only part of the gain. The additional improvement in the full architecture is consistent with three design choices that the baselines lack—canonical ontology normalization, deterministic checking of numeric and temporal constraints, and cross-agent conflict resolution before a recommendation is surfaced. GPT-4 zero-shot prompting and the GPT-4o frontier baseline underperformed substantially without domain-specific graph grounding and multi-agent decomposition.

Operational impact

Over the 12-month period:

- 3804 Patients were screened.
- 23 912 Candidate patient—trial pairs were generated by the AI.
- 17 912 Matches were confirmed as valid after oncologist review.
- Median time-to-recommendation (T1-T0) was <7 days (interquartile range 3-9 days), from initial prescreening (T0) to an oncologist-validated list in the EHR (T1).

Median per-patient screening time fell from ∼120 min (manual) to ∼30 min end-to-end with AI assistance (∼15 min automated processing plus ∼15 min clinician verification and discussion; Figure 3).

- ∼15 Min automated processing.
- ∼15 Min clinician verification and discussion.

At typical staffing levels, this reduces thousands of hours of manual chart review, allowing coordinators to focus on counseling, consent, and protocol adherence.

Extraction fidelity was high for internal EHR data (>95%) and modestly lower for external scanned referrals (∼89%), mainly due to low-resolution biomarker scans. Low-confidence extractions were routed for human verification, preventing upstream noise from propagating into false-positive matches and supporting robustness under documentation drift. This preserved the integrity of clinician-facing recommendations even when OCR degraded.

Fairness and bias

No demographic subgroup exceeded the pre-specified 10-point F1 gap threshold. The largest observed difference was ∼7 points between white and black/African American patients, associated with documentation and care fragmentation (Supplementary Table S4, available at https://doi.org/10.1016/j.esmorw.2026.100706):

- Incomplete comorbidity documentation.
- Fragmented care across multiple health systems.
- Variable note length and detail across sites.

We treated subgroup gaps as equity signals, not only statistical artifacts. Because the system is conservative—flagging missing mandatory criteria as Requires Clinician Judgment or Ineligible rather than hallucinating—missing data reduced automated recall for some patients. Bootstrap comparisons suggested some gaps may be statistically significant, but we primarily interpreted magnitude and direction as equity risk. Black/African American patients had a 12% higher rate of missing structured staging data than white patients, which correlated with lower recall, indicating the disparity was driven mainly by upstream documentation completeness and interoperability rather than model reasoning bias. This links algorithmic fairness to data equity: improving performance for underrepresented groups requires upstream documentation and access interventions.

Error analysis

Common false-positive drivers included (Table 2):

- Ambiguous note language (e.g. 'history of pneumonitis' without clarification of resolution or severity).
- Missing or inconsistently documented comorbidities (e.g. remote myocardial infarction not reflected in structured problem lists).
- Unclear temporal relationships between therapies (e.g. active versus prior immunotherapy lines).

False negatives were often associated with:

- Complex temporal criteria (e.g. 'progression after two prior lines within six months') where approximate date

Table 1. Performance comparison across models on held-out test set (patient—trial pairs)

Model	Sensitivity	Specificity	Precision	F1 score
Multi-agent AI system	0.8375	0.8359	0.8121	0.8246
GPT-4 zero-shot	0.48	0.50	0.47	0.47
GPT-4 CoT baseline	0.67	0.69	0.66	0.67
GPT-4o (matching accuracy)	—	—	—	0.47 (47.00%)
Manual screening	Reference	Reference	Reference	Reference

AI, artificial intelligence; CoT, chain-of-thought.

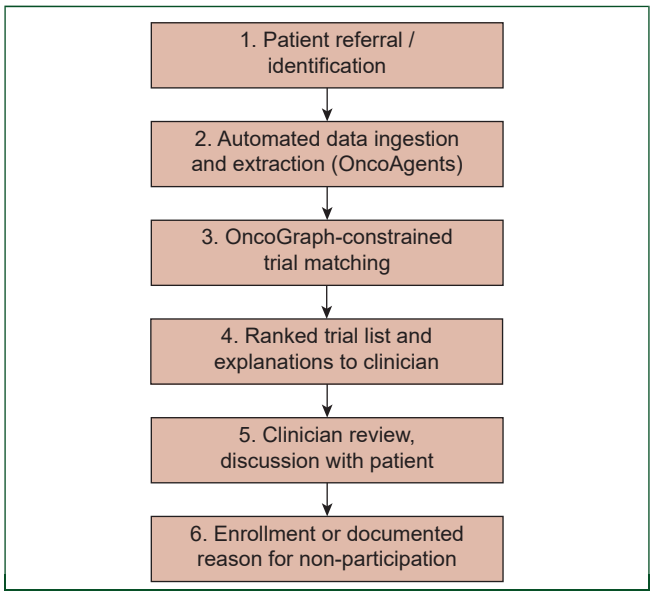


Figure 3. Multi-agent workflow across the patient journey, from referral and automated extraction to clinician review and potential trial enrolment.

ranges or loosely specified timing in notes challenged automated reasoning.
- Rare biomarker combinations not yet fully modeled in OncoGraph or newly emergent molecular targets not represented in the ontologies at ingestion time.
- Subtle or partially scanned biomarker mentions in pathology reports (e.g. low-resolution scanned PDFs with incomplete OCR).

These patterns informed targeted improvements to extraction agents and expanded OncoGraph coverage in subsequent versions.

Ablation analysis

We conducted ablation experiments on the held-out test set (Supplementary Table S3, available at https://doi.org/10.1016/j.esmorw.2026.100706) to estimate the contribution of key components.

Removing OncoGraph grounding ('no-graph' variant) reduced precision and F1 (0.8246 → 0.79) and produced more hallucinated matches that violated numeric or temporal criteria, highlighting the graph's role in preventing logically inconsistent recommendations.

Collapsing extraction and matching into a single LLM call ('single-agent' variant) simplified the pipeline but lowered extraction fidelity and increased unresolved feature—criteria conflicts. F1 fell to 0.78 versus 0.8246 for the full multi-agent system, suggesting decomposition helps maintain consistency. The multi-agent decomposition appears to help the model maintain focus and consistency on extraction before reasoning.

Disabling uncertainty-based triage increased reviewer time by spreading attention across clear and ambiguous cases; focusing review on the highest-uncertainty ∼25% of pairs concentrated oversight where errors were most likely while preserving sensitivity.

Together, these ablations indicate that both graph grounding and agentic decomposition materially contribute to performance and usability.

Calibration and uncertainty triage

Eligibility confidence scores showed near-optimal calibration across typical screening thresholds (Supplementary Table S5; Figure S1, available at https://doi.org/10.1016/j.esmorw.2026.100706), with mild underconfidence at lower bins and modest overconfidence at extremes.

Varying the 'high-uncertainty' threshold traded reviewer workload against sensitivity (Supplementary Figure S2, available at https://doi.org/10.1016/j.esmorw.2026.100706). The selected threshold directed attention to ∼25% of pairs with the highest uncertainty, balancing safety, efficiency, and human-in-the-loop oversight.

DISCUSSION

Why a neuro-symbolic, multi-agent architecture matters

Across tasks, the neuro-symbolic multi-agent architecture outperformed naive and CoT-prompted LLM baselines by grounding eligibility logic in OncoGraph. This is not a

Table 2. Distribution and characterization of AI error modes

Error category	Frequency[a]	Primary driver	Representative example
Complex temporal logic	32%	Difficulty resolving approximate dates or 'fuzzy' intervals in clinical notes against strict trial windows.	Model failed to calculate if 'progression last spring' met the 'progression within 6 months' exclusion criteria.
Missing structured data	28%	Key eligibility criteria (e.g. ECOG status, specific comorbidities) implied but not explicitly documented in the available EHR window.	Patient flagged as 'Ineligible' due to missing ECOG score, despite active hiking hobby mentioned in social history.
Ambiguity and negation	18%	Linguistically ambiguous phrases or complex double negatives that confused the extraction agent.	Phrase 'history of pneumonitis' extracted as active exclusion condition without clarifying resolution status.
Extraction/OCR artifacts	12%	Loss of fidelity in scanned PDF documents or faxed pathology reports.	HER2 status misread from low-resolution scanned pathology report.
Ontology mismatch	10%	Rare biomarkers or novel synonyms not yet mapped in the knowledge graph (OncoGraph).	Specific rare fusion variant not recognized as a child-term of the required parent gene eligibility node.

AI, artificial intelligence; ECOG, Eastern Cooperative Oncology Group; EHR, electronic health record; HER2, human epidermal growth factor receptor 2; OCR, optical character recognition.
[a]Frequencies are derived from the adjudication of false-positive and false-negative discordant pairs in the validation set. 'Missing Structured Data' primarily resulted in false negatives (conservative failures), while 'Ambiguity' often drove false positives.

marginal gain: graph constraints enforce numeric and temporal rules, reduce hallucinated matches, and make reasoning auditable, while agent decomposition keeps extraction and matching focused and consistent. The pattern aligns with Moravec's Paradox[10]: LLMs handle messy narrative understanding but are fragile on Boolean and temporal checks; separating semantic extraction (OncoAgents) from deterministic reasoning (OncoGraph) improved reliability. This division of labor supports clinician review with concise evidence-linked rationales rather than opaque end-to-end prompts.

The central message of Table 1 is therefore not simply that one model scored higher than another, but that performance improved in the same order as structural constraints were added. Zero-shot prompting performed worst because raw clinical notes and raw eligibility text had to be interpreted in a single unconstrained step. CoT prompting improved performance, consistent with the value of explicit intermediate reasoning, but it still relied on latent, non-executable representations of timing, negation, and eligibility exceptions. The full system's further gain to an F1 of 0.82, together with the ablation drops observed when graph grounding or agentic decomposition were removed, suggests that the largest incremental benefit came from externalizing patient state, normalizing it to oncology concepts, and then executing eligibility rules deterministically.

In practical terms, OncoGraph appears to contribute most strongly to precision by blocking logically inconsistent matches, whereas decomposition into extraction, normalization, and matching agents improves recall and reviewer efficiency by reducing omitted clinical features and resolving conflicts across fragmented documentation.

This architecture:

- Constrains reasoning to validated oncology ontologies, reducing hallucinations and logically inconsistent recommendations.
- Decomposes cognitively complex tasks into modular agents focused on extraction, normalization, and reasoning.
- Maintains explicit CoT explanations that support rapid clinician verification and audit.
- Integrates smoothly into existing workflows via human-in-the-loop review and uncertainty-based triage.

We view this as a practical template for deploying agentic AI in safety-critical clinical workflows.

Relation to prior work

Prior multi-agent, knowledge-grounded clinical systems provide important context.[4,11] We extend this approach to broad tumor coverage and a large consecutive prospective cohort, showing that modular agents plus graph grounding can scale in routine practice (3804 patients; 23 912 patient-trial pairs). This broad coverage complements tightly scoped pre-screening programs and stresses robustness to real-world documentation heterogeneity.

These findings address a specific knowledge gap in the trial-matching literature. Prior systems have demonstrated that LLMs, retrieval, and ontology alignment can help identify candidate trials,[3,4,11-13] but many published approaches remain retrospective, disease-limited, or optimized for text similarity and retrieval rather than auditable patient—trial execution over longitudinal records. Our framework differs in combining multimodal EHR extraction, ontology grounding, explicit temporal and exception handling, uncertainty triage, and prospective clinician adjudication within a broad oncology cohort. The contribution is therefore less a claim of universal superiority than a demonstration that neuro-symbolic orchestration can be deployed prospectively at scale with transparent logic and measurable operational benefit.

ClinicalAgent and related knowledge graph CoT/RAG systems[7,12] advocate modular reasoning over structured knowledge, while frameworks like EHRMIND emphasize verifiable, calibrated EHR-grounded methods.[5] Our study adds prospective, equity-aware real-world evidence, including calibration and uncertainty triage.

Direct head-to-head benchmarking against proprietary systems was not feasible, but our metrics align with or exceed published retrieval-augmented screening results in oncology, particularly in reducing hallucinated matches through knowledge graph grounding.

Pipelines such as TRIALSCOPE demonstrate oncology-scale text normalization and ontology alignment.[13] Our work builds on these structuring foundations by closing the loop to patient-level trial recommendations.

Overall, the field is converging on LLMs embedded in structured, verifiable, equity-aware clinical systems—aligned with guidance such as TRIPOD-LLM.[14]

Equity as a first-class performance metric

The ∼7-point F1 deficit for black/African American patients, though below our alert threshold, is clinically important given longstanding underrepresentation in oncology trials. The equity results are encouraging but should not be overinterpreted. No subgroup crossed the pre-specified 10-point F1 alert threshold, yet the ∼7-point gap observed between white and black/African American patients remains clinically meaningful because disparities in trial access are already well documented. The observed error pattern suggests that this gap was driven primarily by upstream data completeness—particularly fragmented care histories, missing staging information, and incompletely documented comorbidities—rather than by qualitatively different reasoning behavior of the model itself.

This distinction is important because improving fairness in this context is as much a documentation and interoperability problem as a model optimization problem. Accordingly, mitigation requires both modeling work

and upstream improvements in clinical documentation completeness.

Our planned mitigation strategies include:

- Fairness-aware model optimization and subgroup-specific calibration.
- Targeted augmentation of training data for underrepresented groups.
- Continuous subgroup performance monitoring and drift detection.
- Integration of claims and real-world data to backfill missing comorbidities and treatment histories.

Future multi-site deployments should treat equity metrics as co-primary outcomes and include explicit equity guardrails.

Distribution shift and external validation

This evaluation occurred within a single health system with relatively consistent documentation norms. External validation across diverse EHR vendors, sites, and patient populations is essential before broad deployment to mitigate distribution shift. Our single-system prospective evaluation demonstrates that the architecture can function under real clinical documentation conditions, but distribution shift is likely to arise both from local documentation conventions and from evolving trial portfolios, biomarker requirements, and protocol language. For this reason, site-specific calibration, local ontology mapping, and routine drift monitoring should be considered ongoing operational requirements rather than one-time validation exercises.

Our ongoing work includes:

- Pilot implementations in community sites with different EHR vendors and note styles.
- Site-specific calibration and adaptation of extraction agents (e.g. mapping local laboratory codes and templates).
- Robustness testing against variable note structures and imaging/pathology formats.

We will report performance and fairness across OncoGraph and model versions to characterize drift and adaptation effectiveness.

Technical debt and knowledge graph maintenance

Maintaining OncoGraph is essential but nontrivial: ontologies evolve and trial portfolios change, requiring ongoing curation, conflict resolution, and monitoring.

We tracked maintenance effort to keep OncoGraph current: ∼20 h/month of expert curator time plus ∼4 h/month of senior oncologist oversight. This fixed cost functions as a 'price of safety'—a deterministic logic layer that can be centrally audited and updated, unlike pure LLM systems that require continual re-prompting or retraining to correct hallucinations. Knowledge graph maintenance therefore represents a form of intentional technical debt. While ingestion and mapping are partly automated, ongoing human review remains necessary as ontologies, biomarkers, and trial portfolios evolve. Pure LLM pipelines may appear lighter weight but often relocate this burden into repeated prompt tuning, silent failure modes, or *post hoc* chart review.

Future work will:

- Quantify the operational burden of graph maintenance.
- Explore semi-automated ontology reconciliation tools.
- Assess performance stability across OncoGraph versions.

We consider this maintenance burden justified by the interpretability and safety gains.

Regulatory and human factors considerations

Given its role as clinical decision support, the platform may qualify as Software as a Medical Device (SaMD). From a human factors and regulatory perspective, the system is most defensible when treated as clinician-facing decision support rather than autonomous eligibility determination. The most important safeguards are not only model accuracy but also:

- Evidence-linked rationales supporting clinician verification.
- Conservative handling of missing mandatory criteria.
- Uncertainty-based escalation for ambiguous cases.
- Version-controlled updates to models and graph logic.

These features align with emerging trial protocol guidance (SPIRIT-AI; SPIRIT 2025).[15,16] To mitigate automation bias, deployments should include training, audits of false negatives, and guardrails when trial portfolios or criteria change.

Limitations and future directions

Our evaluation reports identification efficiency—time, accuracy, calibration, and fairness—but not downstream endpoints such as consent, enrollment, or trial completion. We distinguish between 'identification efficiency' and 'enrollment success': accurate matching is necessary but not sufficient because logistical, socioeconomic, and consent barriers persist. Reported gains primarily reduce the burden of identifying eligible patients; future work will measure conversion from AI-matched to consented across the full enrollment funnel.

Head-to-head comparison with every proprietary matching system was not feasible, and ablation analyses reflect our specific implementation. The recurring failure modes observed in the LLM-only baselines, however—hallucinated criteria, inconsistent temporal reasoning, and vulnerability to documentation gaps—suggest that the value of symbolic constraint is structural rather than model-specific.

Future work should therefore focus on multi-site validation, version-to-version drift reporting, and end-to-end measurement of how improved matching translates into actual trial enrollment.

CONCLUSIONS

In a prospective evaluation of 3804 consecutive patients, processing 157 367 clinical pages and >23 000 candidate patient—trial pairs, a neuro-symbolic multi-agent platform improved trial-matching accuracy, throughput, and timeliness while preserving clinician oversight and an equity-aware deployment framework.

By outperforming zero-shot, CoT, and frontier GPT-4o models and reducing screening time fourfold, the system offers a practical path for responsible AI integration into oncology research. The findings provide a transparent, calibrated, and equity-aware foundation for future collaborative validation studies and illustrate how agentic LLM components can be constrained by symbolic structure for safety-critical deployment.

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DISCLOSURE

ALB is co-founder and Chief Medical and AI Officer of Massive Bio, Inc., and reports equity ownership in the company. He also serves as System-Wide Chief of Hematology and Oncology at St. Luke's University Health Network. SK is co-founder and Chief Executive Officer of Massive Bio, Inc., and reports equity ownership. ET, JPNM, and DG are employees of Massive Bio, Inc., and report no additional conflicts of interest. All other authors have declared no conflicts of interest. The authors declare that no funding source influenced the study design, data collection, analysis, interpretation, writing of the report, or the decision to submit the article for publication.

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